

MH1101 Chapter 2: Applications of Integrations

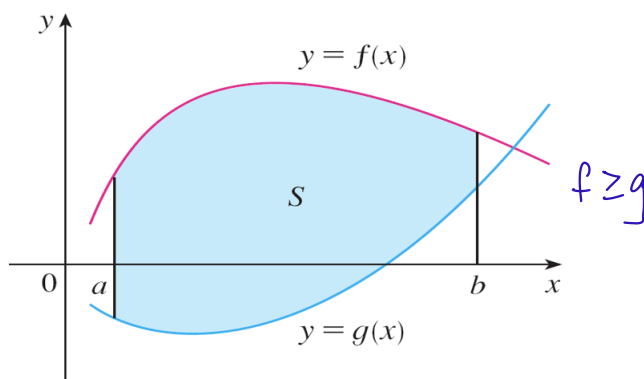
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2.1 Area Between Curves

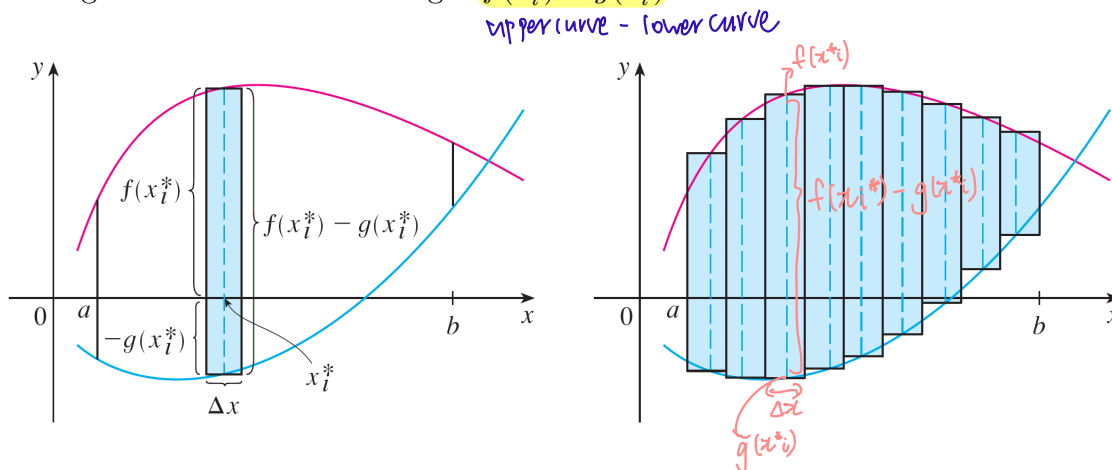
Reference: Stewart Calculus 8E, Section 5.1.

Goal: Use definite integrals to find areas of regions bounded between two curves.

Consider the region S that lies between two curves $y = f(x)$ and $y = g(x)$ and between the vertical lines $x = a$ and $x = b$, where f and g are continuous functions and $f(x) \geq g(x)$ for all $x \in [a, b]$.



We divide S into n strips of equal width, and then approximate the l -th strip by a rectangle with base Δx and height $f(x_l^*) - g(x_l^*)$.



The Riemann sums

$$\sum_{i=1}^n (f(x_i^*) - g(x_i^*)) \Delta x$$

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is an approximation to what we intuitively think of as the ~~area of~~ ^{of} ~~S~~ ^S.

Indeed, this approximation gets better and better as $n \rightarrow \infty$, and we define the area of S to be

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \overbrace{(f(x_i^*) - g(x_i^*))}^{h(x_i^*)} \Delta x$$

$h = f - g$
 $\int_a^b h(x) dx = \int_a^b (f - g) dx$
equals

We recognise this as a definite integral of $f - g$. Thus, we have the following formula of area between two curves:

Theorem 1. (Area between two curves)

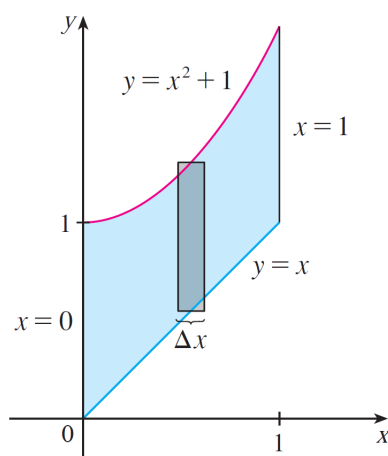
The area A of the region bounded by the curves $y = f(x)$, $y = g(x)$, and the vertical lines $x = a$, $x = b$, where f and g are continuous and $f(x) \geq g(x)$ for all x in $[a, b]$ is

$$A = \int_a^b (f(x) - g(x)) dx.$$

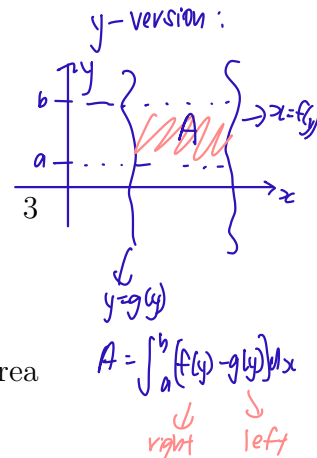
Always upper - lower
upper curve *lower curve*

Example 2.1. Find the area A of the region bounded above by $y = x^2 + 1$, bounded below by $y = x$, and bounded on the sides by the lines $x = 0$ and $x = 1$.

Solution. Notice $y = x^2 + 1$ is above $y = x$ on $[0, 1]$:



We apply the formula with $f(x) = x^2 + 1$, $g(x) = x$:

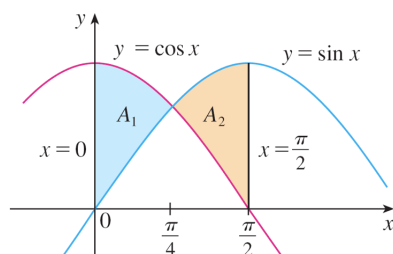


$$\begin{aligned}
 A &= \int_0^1 \overbrace{(x^2 + 1 - x)}^{\text{upper}} \underbrace{}_{\text{lower}} dx \\
 &= \left[\frac{x^3}{3} + x - \frac{x^2}{2} \right]_0^1 \\
 &= \frac{5}{6}.
 \end{aligned}$$

□

Example 2.2. Find the area A of the region bounded by the curves $y = \sin x$, $y = \cos x$, and the lines $x = 0$ and $x = \pi/2$.

Solution. The region is sketched below:



*upper and lower curve
are different at different
intervals!

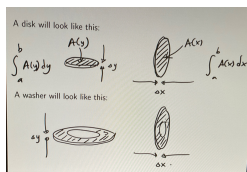
↳ cannot apply formula to
entire interval, need to
split it up

Let $f(x) = \sin x$, $g(x) = \cos x$.

Note: $g(x) \geq f(x)$ on $[0, \pi/4]$, whereas $f(x) \geq g(x)$ on $[\pi/4, \pi/2]$. Using the area formula, we have

$$\begin{aligned}
 A &= \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx \\
 &= [\sin x + \cos x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{\pi/2} \\
 &= \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 0 - 1 \right) + \left(-0 - 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) \\
 &= 2\sqrt{2} - 2.
 \end{aligned}$$

□



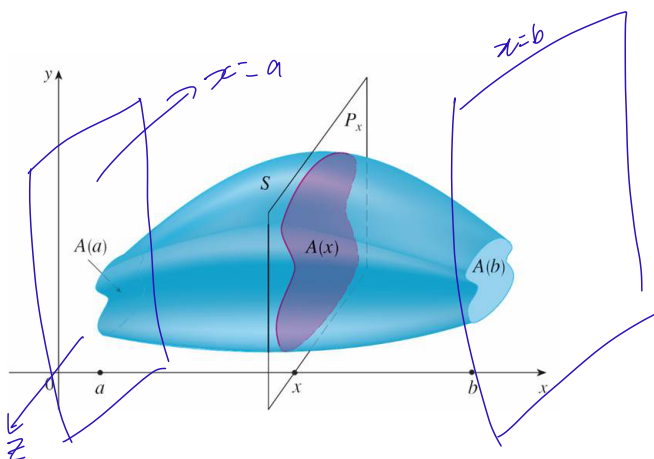
2.2 Volumes by Disk-Washer Method

Reference: Stewart Calculus 8E, Section 5.2.

Goal: Use definite integrals to find volumes of solids generated by revolving regions about vertical or horizontal lines. By “slicing” a solid of revolution (perpendicularly to the axis of rotation), we find the volume by integrating the areas of the “disks” or “washers” corresponding to the cross-sectional areas.

First, let us define what do we mean by the volume of a solid in general.

Consider the following solid S . We assume the solid is placed along the x -axis, between $x = a$ and $x = b$.



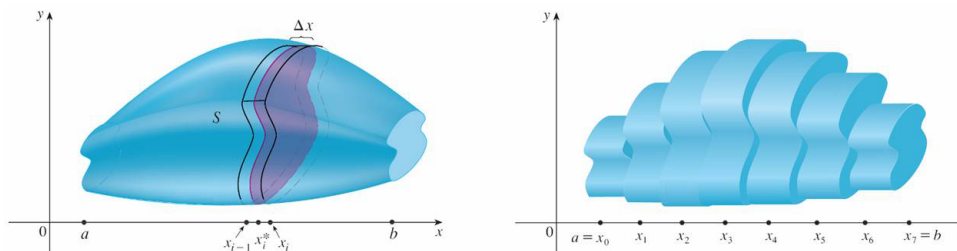
We approximate the volume $V(S)$ of S as follows:

- Divide S into n “slabs” of equal width Δx . (Think of slicing a loaf of bread.)
- The volume of each i -th slab is approximately

$$A(x_i^*) \Delta x$$

↓ sample point
very small $\Delta x \rightarrow$ so small that cross sectional area is the same

where $A(x_i^*)$ is the area of the cross-section of S when we slice it with a vertical plane.



- Adding these slabs, the volume $V(S)$ is approximately

$$V(S) \approx \sum_{i=1}^n A(x_i^*) \Delta x.$$

- Taking limit as $n \rightarrow \infty$, we have

$$V(S) = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i^*) \Delta x.$$

We recognise the limit as the definite integral of $A(x)$ over $[a, b]$. So we have arrived at the following definition of volume.

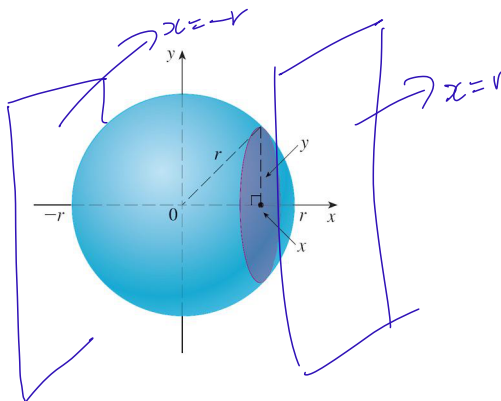
Definition 1. (Volume of Solid)

Let S be a solid that lies between $x = a$ and $x = b$. If the **cross-sectional area** of S in the plane P_x through x and perpendicular to the x -axis is $A(x)$, then the **volume** of S is

$$V(S) = \int_a^b A(x) dx.$$

Example 2.3. Show that the volume V of a sphere of radius r is $\frac{4}{3}\pi r^3$.

Solution. Place the sphere so that its centre is at the origin:

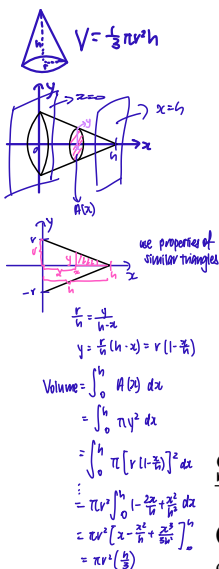


The cross-sectional area $A(x)$ is $\pi(r^2 - x^2)$ by the Pythagorean Theorem.

Using the definition of volume, with $a = -r$ and $b = r$, we have

eqn of circle: $x^2 + y^2 = r^2$
 $y^2 = r^2 - x^2$

Extra Example:



$$\begin{aligned} V &= \int_{-r}^r A(x) dx = \int_{-r}^r \pi y^2 dx \\ &= \int_{-r}^r \pi (r^2 - x^2) dx \quad \text{radius of disc} \\ &= \pi \left[r^2 x - \frac{x^3}{3} \right]_{-r}^r \\ &= \frac{4}{3} \pi r^3. \end{aligned}$$

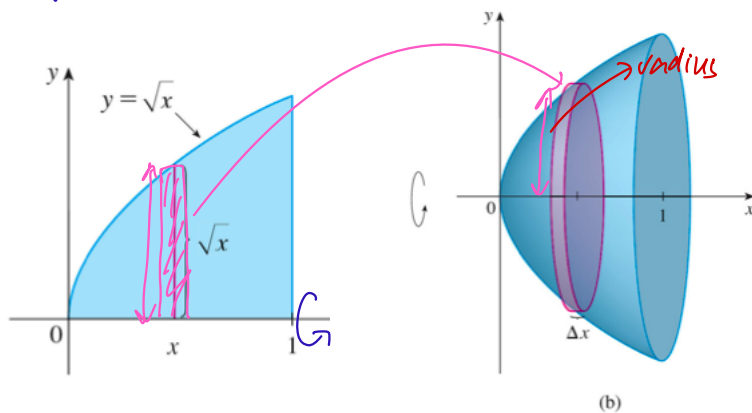
□

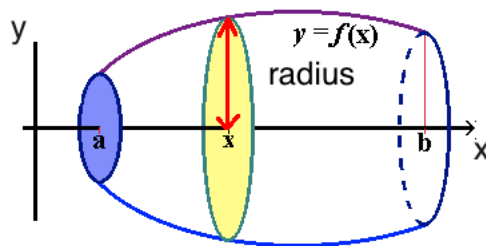
Solids of Revolution.

One common way of generating a solid is by revolving a plane region about a line. Such solids are called **solids of revolution**.

Suppose we have a plane region bounded between a curve $y = f(x)$ and the x -axis (or the y -axis). A solid of revolution is generated by revolving this plane region about the x -axis (or the y -axis). The **radius** of this solid of revolution is the distance between the curve and the axis of rotation.

won't have slanted lines





In general, we can calculate the volume of a solid of revolution by using the defining formula

$$V = \int_a^b A(x) dx \quad (\text{slicing perpendicularly to a horizontal axis of rotation})$$

$$V = \int_c^d A(y) dy \quad (\text{slicing perpendicularly to a vertical axis of rotation}).$$

The cross-sectional area $A(x)$ or $A(y)$ is found by using the following **Disk-Washer** method:

- **The Disk Method:**

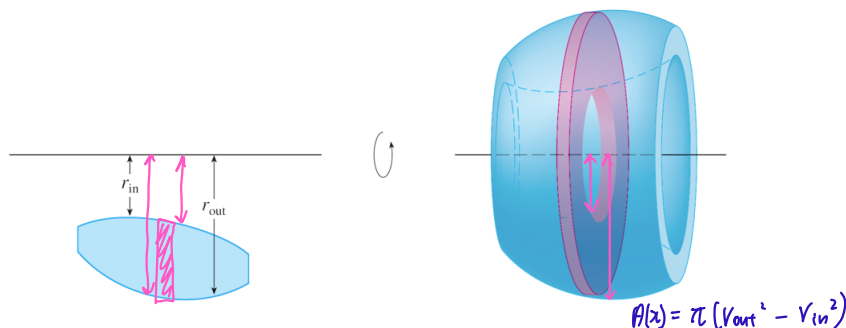
If the cross-section is a disk, we find the radius of the disk (in terms of x or y) and the cross-sectional area is

$$A = \pi(\text{radius})^2.$$

- **The Washer Method:**

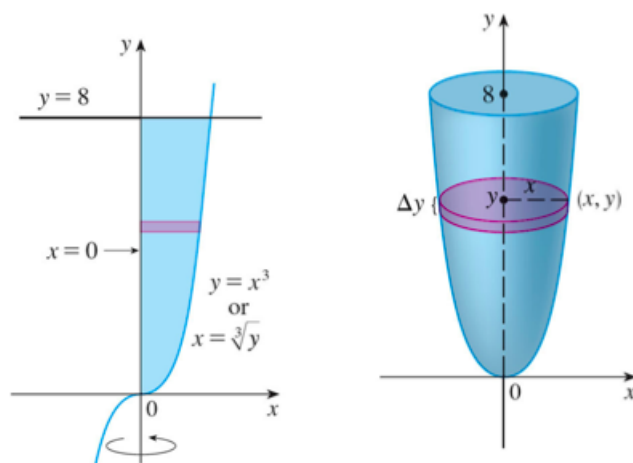
If the cross-section is a washer, we find the inner radius r_{in} and outer radius r_{out} from a sketch, and compute the cross-sectional area by subtracting the area of the inner disk from the area of the outer disk:

$$A = \pi(\text{outer radius})^2 - \pi(\text{inner radius})^2.$$



Example 2.4. Find the volume of the solid obtained by rotating the region bounded by $y = x^3$, $y = 8$, and $x = 0$ about the y -axis.

Solution.



If we slice the solid at height y perpendicular to the y -axis, we obtain a circular disk of radius $x = y^{1/3}$, where $0 \leq y \leq 8$. So the area of a cross section through y is

$$A(y) = \pi x^2 = \pi (y^{1/3})^2 = \pi y^{2/3}.$$

rotation about y axis
as we slice horizontally

Since the solid lies between $y = 0$ and $y = 8$, its volume is (by the Disk Method)

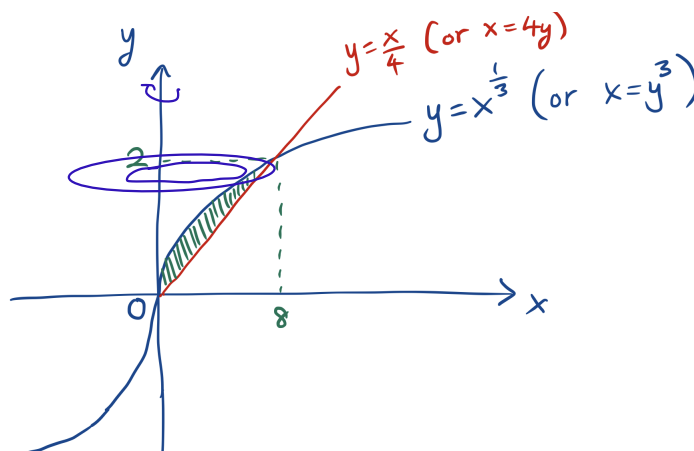
$$V = \int_0^8 A(y) dy = \int_0^8 \pi y^{2/3} dy = \pi \left[\frac{3}{5} y^{5/3} \right]_0^8 = \frac{96\pi}{5}.$$

π(radius)²

□

Example 2.5. Determine the volume of the solid generated by rotating the region bounded by $y = x^{1/3}$, and $y = \frac{x}{4}$ that lies in the **first quadrant** about the y -axis.

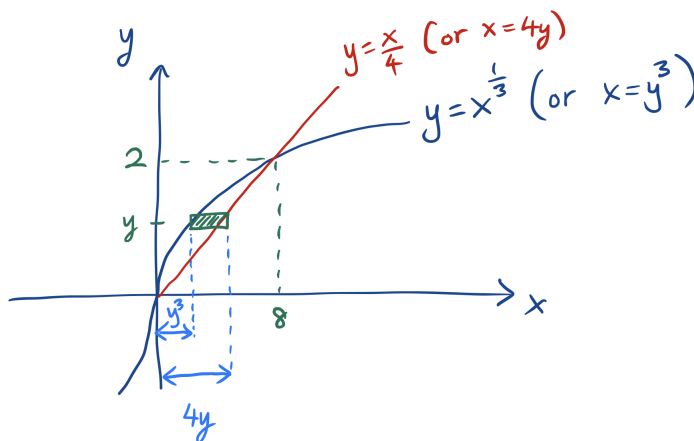
Solution. The cross section is cut perpendicular to the axis of rotation (y -axis) and it is a horizontal washer, where $0 \leq y \leq 2$.



We need to write our functions in the form $x = f(y)$.

$$y = x^{1/3} \implies x = y^3.$$

$$y = \frac{x}{4} \implies x = 4y.$$



The inner radius is $x = y^3$, and the other radius is $x = 4y$. Thus, the cross-sectional area is

$$A(y) = \pi(\overset{\text{Vout}}{(4y)^2} - \overset{\text{VIn}}{(y^3)^2}) = \pi(16y^2 - y^6).$$

The two curves $x = 4y$ and $x = y^3$ intersect when $4y = y^3$, whence

$$y^3 - 4y = 0,$$

$$y(y^2 - 4) = 0,$$

that is either $y = 0$ or $y = \pm 2$. Since the solid lies in the first quadrant, it must lie between $y = 0$ and $y = 2$.

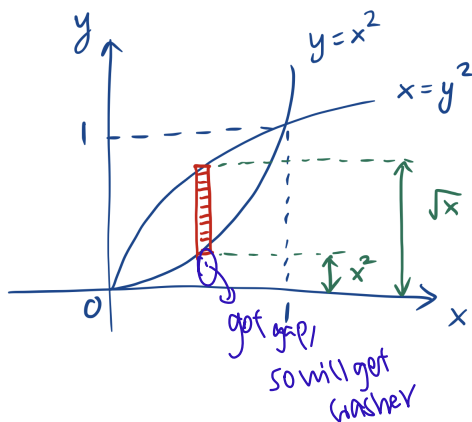
By the Washer Method, the volume is

$$\begin{aligned} V &= \int_0^2 A(y) dy = \int_0^2 \pi(16y^2 - y^6) dy \\ &= \pi \left[\frac{16y^3}{3} - \frac{y^7}{7} \right]_0^2 \\ &= \frac{512\pi}{21}. \end{aligned}$$

□

Example 2.6. Find the volume of the solid obtained by rotating the region bounded by the curves $y = x^2$ and $x = y^2$ in the **first quadrant** about the horizontal line $y = 0$.

Solution. The cross section is cut perpendicular to the axis of rotation ($y = 0$) and it is a vertical washer, where $0 \leq x \leq 1$.



- The outer radius is the distance from $y = 0$ to $y = \sqrt{x}$.
- The inner radius is the distance from $y = 0$ to $y = x^2$.

So the volume is

$$\begin{aligned} V &= \int_0^1 \pi(\overset{\text{r}_{\text{out}}}{\sqrt{x}})^2 - \pi(\overset{\text{r}_{\text{in}}}{x^2})^2 dx \\ &= \pi \int_0^1 x - x^4 dy \\ &= \pi \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1 \\ &= \frac{3\pi}{10}. \end{aligned}$$

□

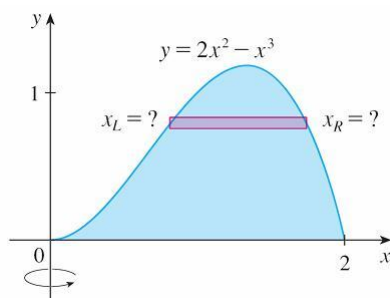
2.3 Volumes by Cylindrical Shells

Reference: Stewart Calculus 8E, Section 5.3.

Goal: Find volume of solid of revolution using the method of cylindrical shells. This is created by “slicing” parallel to the axis of rotation.

Let us motivate the method using the following example.

Consider the problem of finding the volume of the solid obtained by rotating about the y -axis the region bounded by $y = 2x^2 - x^3$ and $y = 0$ (see figure below).

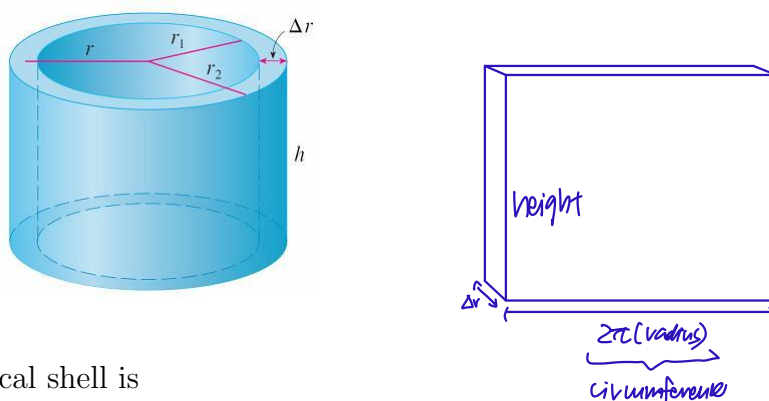


If we slice perpendicular to the y -axis, we get a washer. But to compute the inner radius and the outer radius of the washer, we'd have to solve the cubic equation $y = 2x^2 - x^3$ for x in terms of y ; that's not easy.

Fortunately, there is a method, called the method of **cylindrical shells**, that is easier to use in such a case.

The figure below shows a cylindrical shell with

- inner radius r_1
- outer radius r_2
- height h



The volume V of this cylindrical shell is

$$\begin{aligned}
 V &= (\text{Volume of the outer cylinder}) - (\text{Volume of the inner cylinder}) \\
 &= \pi r_2^2 h - \pi r_1^2 h \\
 &= \pi h(r_2 - r_1)(r_1 + r_2) \\
 &= 2\pi h(r_2 - r_1) \frac{r_1 + r_2}{2}.
 \end{aligned}$$

Let $\Delta r = r_2 - r_1$ (the thickness of the shell) and $r = \frac{1}{2}(r_1 + r_2)$ (the average radius of the shell), then the volume of the cylindrical shell becomes

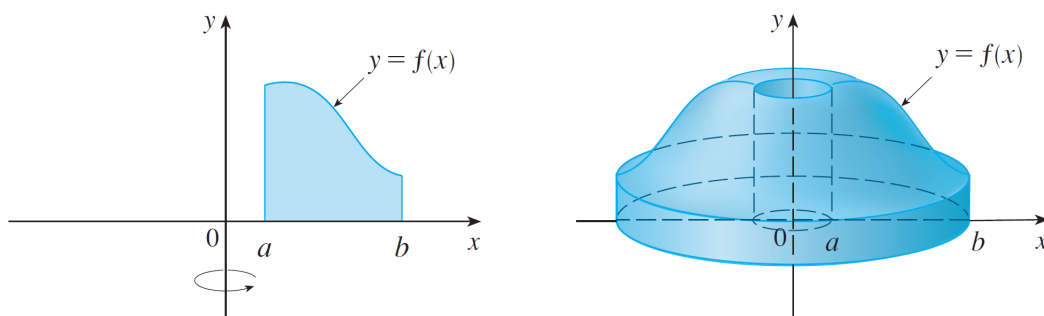
$$V = 2\pi r h \Delta r,$$

and it can be remembered as

$$V = \underbrace{2\pi r}_{\text{circumference}} \cdot \underbrace{h}_{\text{height}} \cdot \underbrace{\Delta r}_{\text{thickness}}.$$

Let us now derive a formula for volume using cylindrical shells.

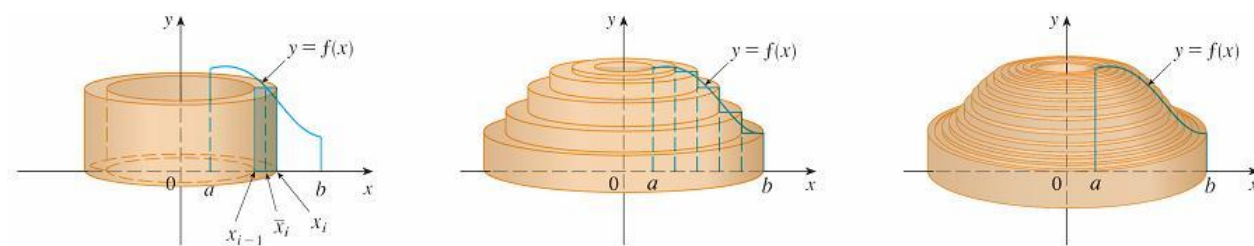
Consider the solid S obtained by rotating a region bounded by $y = f(x)$ (where $f(x) \geq 0$), $x = a$, and $x = b$, where $0 \leq a < b$, about the y -axis.



The idea is to **slice it parallel** to the axis of revolution. Each i -th (vertical) strip generates a cylindrical shell (when revolved about the axis) whose volume is

$$V_i = 2\pi \bar{x}_i f(\bar{x}_i) \Delta x,$$

where $\bar{x}_i$ is the average radius of the cylindrical shell, $f(\bar{x}_i)$ is the height of the cylindrical shell, and Δx is the thickness of the cylindrical shell.



The volume V of S can be approximately given by

$$V \approx \sum_{i=1}^n V_i = \sum_{i=1}^n 2\pi \bar{x}_i f(\bar{x}_i) \Delta x. \quad \rightarrow \text{Riemann sum}$$

This approximation becomes exact as $n \rightarrow \infty$. So by the definition of a definite integral, we have

$$V = \int_a^b 2\pi x f(x) dx.$$

A similar argument works for solids revolving about the x -axis.

In general, we have the following formula for volume using cylindrical shells.

Theorem 2. (Volume by Cylindrical Shells)

- (a) The volume of a solid obtained by revolving a region (bounded between $x = a$ and $x = b$) about a vertical line is

$$V = \int_a^b 2\pi(\text{radius})(\text{height}) dx.$$

Here, the radius and the height of the cylindrical shells are expressed in terms of x .

- (b) The volume of a solid obtained by revolving a region (bounded between $y = c$ and $y = d$) about a horizontal line is

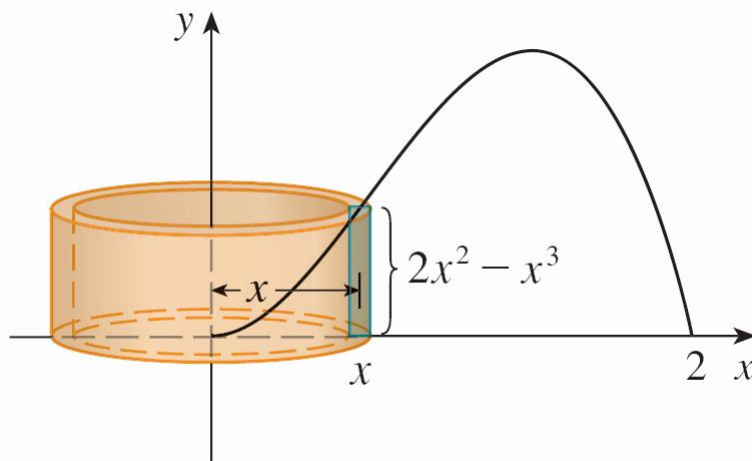
$$V = \int_c^d 2\pi(\text{radius})(\text{height}) dy.$$

Here, the radius and the height of the cylindrical shells are expressed in terms of y .

Example 2.7. Find the volume of the solid obtained by rotating about the y -axis the region bounded by $y = 2x^2 - x^3$ and $y = 0$.

Solution. We slice the solid using vertical strips. From the figure below, a typical cylindrical shell has

- radius = x
- height = $2x^2 - x^3$



So, by the shell method, the volume is

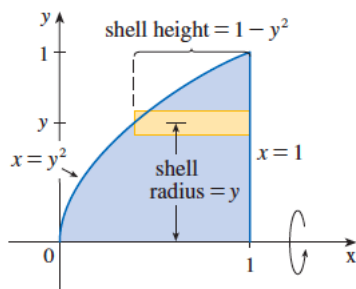
$$\begin{aligned} V &= \int_0^2 2\pi(x)(2x^2 - x^3) dx \\ &= 2\pi \int_0^2 (2x^3 - x^4) dx \\ &= 2\pi \left[\frac{x^4}{2} - \frac{x^5}{5} \right]_0^2 \\ &= 2\pi \left(8 - \frac{32}{5} \right) \\ &= \frac{16\pi}{5}. \end{aligned}$$

□

Example 2.8. Use cylindrical shells to find the volume of the solid obtained by rotating about the x -axis the region under the curve $y = \sqrt{x}$ and above the x -axis from $x = 0$ to $x = 1$.

Solution. We slice the solid using horizontal strips. From the figure below, a typical cylindrical shell has

- radius = y
- height = $1 - x = 1 - y^2$



IC Method:

$$\int_0^1 \pi y^2 dx$$

$$= \int_0^1 \pi x dx$$

$$= \pi \left[\frac{x^2}{2} \right]_0^1$$

$$= \frac{\pi}{2}$$

So, by the shell method, the volume is

$$\begin{aligned} V &= \int_0^1 2\pi(y)(1 - y^2) dy \\ &= 2\pi \int_0^1 (y - y^3) dy \\ &= 2\pi \left[\frac{y^2}{2} - \frac{y^4}{4} \right]_0^1 = \frac{\pi}{2}. \end{aligned}$$

□

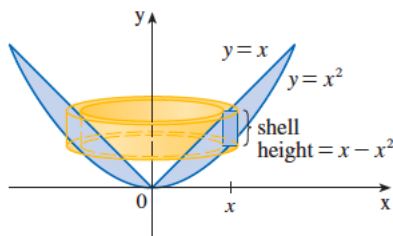
Example 2.9. Find the volume of the solid obtained by rotating about the y -axis the region between $y = x$ and $y = x^2$.

Solution. The region bounded by $y = x$ and $y = x^2$ lies in the first quadrant. The two curves intersect when

$$x = x^2 \implies x^2 - x = x(x - 1) = 0.$$

So the two curves intersect at two points: $(0, 0)$, $(1, 1)$.

A typical cylindrical shell is shown in the figure below.



Extra:
 Change axis of rotation to $x = 2$
 radius = $2 - x$
 $\rightarrow$ don't change region!
 $V = \int_0^1 2\pi (2-x)(x-x^2) dx$
 Change axis of rotation to $x = -5$
 radius = $x - (-5) = x + 5$
 Volume = $\int_0^1 2\pi (x+5)(x-x^2) dx$

A typical shell has

- radius x
- circumference $2\pi x$
- height $x - x^2$

So the volume is

$$\begin{aligned} V &= \int_0^1 (2\pi x)(x - x^2) dx \\ &= 2\pi \int_0^1 (x^2 - x^3) dx \\ &= 2\pi \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 \\ &= \frac{\pi}{6}. \end{aligned}$$

* cylindrical shell not possible ALL the time
 similar problem as disk washer

□